All New Matrix Groups and Formulas

Consolidated theorem and verification compendium

Lambda Distributions workspace

20 July 2026

Abstract

This document is the cross-family index for the matrix-group results developed in the workspace. It records every new group family, the formula actually proved or corrected, its finite/stable/probabilistic status, and the principal verification evidence. The detailed proofs, notebooks, and group-specific PDFs remain in `lambda_distributions/proofs` and `lambda_distributions/dists`. The common object is the generalized Molien or sigma-MGF series, but the compendium also includes two-sided unitary series, nonuniform measures, heat kernels, and coefficientwise limits.

Global outcome. Exact finite-group formulas were checked by explicit matrices, Reynolds ranks, orbit counts, or independent coefficient recurrences. Compact-group formulas were checked by exact Weyl constant terms, Lie-algebra kernels, or seeded Haar diagnostics. Claims requiring a correction or a qualification are marked explicitly; no noncompact group is assigned a normalized Haar probability.

Contents

1 Common notation and master identities

For a representation $\rho : G \to GL(V)$ and a partition $\tau = (\tau_1, \dots, \tau_s)$, set

$$W_\tau(V) = \bigotimes_{a=1}^{s} \mathrm{Sym}^{\tau_a} V, \qquad |\tau| = \sum_a \tau_a.$$

For finite G, the Reynolds projector gives the exact identity

$$\boxed{\mathcal{S}_{G,V}(\mathbf{t}) = \frac{1}{|G|}\sum_{g\in G}\prod_i \det(I - t_i\rho(g))^{-1} = \sum_\tau \dim W_\tau(V)^G\, m_\tau.} \tag{1}$$

Equivalently,

$$\mathcal{S}_{G,V} = \frac{1}{|G|}\sum_g \exp\left(\sum_{i,r\geq 1}\frac{\mathrm{Tr}(\rho(g)^r)}{r}t_i^r\right).$$

For compact G, normalized Haar integration replaces the finite average:

$$\boxed{\mathcal{S}_{G,V} = \int_G \prod_i \det(I - t_i\rho(g))^{-1}dg = \sum_\lambda \dim(\mathbb{S}_\lambda V)^G s_\lambda = \sum_\tau \dim W_\tau(V)^G m_\tau.} \tag{2}$$

If T is a maximal torus with representation weights μ and root system Φ, Weyl integration gives

$$\mathcal{S}_{G,V} = \frac{1}{|W|}\,\mathrm{CT}_z\left[\prod_{\alpha>0}(1-z^{\alpha})(1-z^{-\alpha})\prod_{i,\mu}(1-t_iz^{\mu})^{-m(\mu)}\right]. \tag{3}$$

For a probability measure ν that is not Haar, the coefficient is instead the trace of its averaging operator:

$$[m_\tau]\mathcal{S}_{\nu,V} = \mathrm{Tr}\left(\int_G \rho_{W_\tau}(g)d\nu(g)\right). \tag{4}$$

Thus nonuniform coefficients need not be nonnegative integers.

2 Finite abelian, monomial, dihedral, and wreath families

2.1 Finite abelian zero-weight theorem

For $A = \prod_{\ell=1}^{r} C_{m_\ell}$ and $V = \bigoplus_{j=1}^{d} \mathbb{C}_{w_j}$ with weight matrix $W = (w_{\ell j})$,

$$\boxed{\dim W_\tau(V)^A = \#\left\{q_{bj}\ge 0 : \sum_j q_{bj} = \tau_b,\quad \sum_{b,j} w_{\ell j}q_{bj} \equiv 0 \pmod{m_\ell}\ \forall \ell\right\}}. \tag{5}$$

This is simultaneously a lattice count, a finite Fourier projection, and the coefficient of the diagonal Molien average.

2.2 Restricted monomial groups and $G(r,p,n)$

For $G(r,p,n)$ on $\mathbb{C}^n$, $p \mid r$, the exact finite-n series is

$$\boxed{\mathcal{S}_{G(r,p,n)} = [u^n]\sum_{q=0}^{p-1}\mathrm{Exp}_\sigma\left(u\sum_{j\ge 0} h_{qr/p+jr}\right)}. \tag{6}$$

This contains the Weyl groups $B_n/C_n = G(2,1,n)$ and $D_n = G(2,2,n)$. More generally, for a code $A \le (\mathbb{Z}/r)^n$ stable under $H \le S_n$, the coefficient for $D(A) \rtimes H$ is the average over H of cycle-label assignments whose phase-exponent vector lies in the dual code $A^\perp$.

2.3 Wreath products

For a finite matrix group $H \le GL(V)$ acting blockwise on $V^{\oplus n}$,

$$\boxed{\sum_{n\ge 0} u^n \mathcal{S}_{H\wr S_n}(\mathbf{t}) = \exp\left(\sum_{r\ge 1}\frac{u^r}{r}\mathcal{S}_H(t_1^r, t_2^r, \ldots)\right)}. \tag{7}$$

The coefficientwise stable series is

$$\mathcal{S}_{H\wr S_\infty} = \mathrm{Exp}_\sigma(\mathcal{S}_H - 1), \qquad \mathcal{S}_{H\wr S_n} \equiv \mathcal{S}_{H\wr S_\infty} \pmod{\deg > n}.$$

The key block-cycle identity is $\det(I - tg_C) = \det(I - t^{|C|}h_C)$.

2.4 Generalized dihedral and dicyclic groups

For $\mathrm{Dih}(A) = A \rtimes C_2$, the irreducibles are the two extensions of every self-inverse character and the two-dimensional $\mathrm{Ind}_A^G \chi$ for pairs $\{\chi, \chi^{-1}\}$. Substitution of their diagonal/reflection spectra in (1) is the master formula for arbitrary direct sums. All 33 generalized-dihedral checks passed.

For the dicyclic group Dic_m and its two-dimensional irreducibles ρ_k, the series is the average of the $2m$ diagonal spectra $(\zeta^{kj}, \zeta^{-kj})$ and the $2m$ antidiagonal spectra. The coefficient form is exactly $\dim W_\tau(\rho_k)^{\mathrm{Dic}_m}$; direct Reynolds, character, and weight calculations agree. The regular-representation formula for any finite group is

$$\boxed{\mathcal{S}_{G,\mathrm{Reg}} = \sum_{d\geq 1} \frac{N_d(G)}{|G|} \operatorname{Exp}_\sigma\left(\frac{|G|}{d} p_d\right),} \tag{8}$$

where $N_d(G)$ counts elements of order d.

3 Symmetric, alternating, cube, and permutation actions

3.1 Exact S_n and A_n formulas

If $\lambda = 1^{m_1} 2^{m_2} \cdots \vdash n$, then the natural permutation series is

$$\boxed{\mathcal{S}_{S_n} = \sum_{\lambda\vdash n} \frac{1}{z_\lambda} \prod_{i,d} (1 - t_i^d)^{-m_d(\lambda)}.} \tag{9}$$

For A_n, retain the even cycle types and double their S_n weights; the extra orbit-splitting term appears only for all-distinct labels. In stable range the A_n and S_n coefficients agree once two unused labels remain.

For the sign representation,

$$\mathcal{S}_{S_n,\mathrm{sgn}} = \tfrac{1}{2}\left[\prod_i (1-t_i)^{-1} + \prod_i (1+t_i)^{-1}\right].$$

For the standard module V_n, removing the trivial line gives

$$\mathcal{S}_{S_n,V_n} = \sum_{\lambda\vdash n} \frac{1}{z_\lambda} \prod_i \left[(1-t_i) \prod_{d\in\lambda} (1-t_i^d)^{-1}\right], \qquad \mathcal{S}_{S_n,V_n} \equiv \operatorname{Exp}_\sigma(h_2 + h_3 + \cdots) \pmod{\deg > n}.$$

3.2 Universal permutation-action formula

For $G \curvearrowright X$ and vertex-cycle counts $c_d(g;X)$,

$$\boxed{\mathcal{S}_{G,X} = \frac{1}{|G|} \sum_g \prod_{i,d} (1-t_i^d)^{-c_d(g;X)}, \quad [m_\tau]\mathcal{S}_{G,X} = \#\left(G\backslash \prod_j \operatorname{Sym}^{\tau_j} X\right).} \tag{10}$$

Cycles may be reconstructed from fixed points by

$$\boxed{c_d(g;X) = \frac{1}{d} \sum_{r\mid d} \mu(d/r)\, |\operatorname{Fix}_X(g^r)|.} \tag{11}$$

This single template covers Johnson k-subsets, Hamming schemes $S_q \wr S_D$, Grassmann schemes $\mathrm{Gr}_q(N,k)$, polar and dual-polar spaces, finite linear/projective/flag actions, and the projective odd-symplectic actions below.

For S_n on k-subsets, with $A_j(\sigma)$ the coordinate cycle counts,

$$\boxed{|\operatorname{Fix}(\sigma^r)| = [u^k] \prod_{j=1}^{kr} (1 + u^{j/\gcd(j,r)})^{\gcd(j,r) A_j(\sigma)}.} \tag{12}$$

For Hamming words, fixed words are the product of the numbers of alphabet symbols fixed by the label product around each coordinate cycle. Grassmann and polar actions replace this by the count of invariant subspaces, with the isotropic/singular predicate in the polar case.

3.3 The hyperoctahedral group on the binary cube

For $B_n = \mathbb{F}_2^n \rtimes S_n$ acting affinely on $X_n = \mathbb{F}_2^n$, conditional on $\pi \in S_n$ with $C(\pi)$ coordinate cycles,

$$F_n(b,\pi) = |\operatorname{Fix}(b+\pi\cdot)| = \begin{cases} 2^{C(\pi)}, & \text{with probability } 2^{-C(\pi)}, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore

$$\boxed{\mathbb{E}[F_n^m] = \frac{(2^{m-1})^{\overline{n}}}{n!} = \binom{n+2^{m-1}-1}{n}.} \tag{13}$$

Signed coordinate-cycle types (a_ℓ, b_ℓ), their exact weights $\prod_\ell ((2\ell)^{a_\ell+b_\ell} a_\ell! b_\ell!)^{-1}$, the fixed-power law, and (11) give the full finite-n cube series. In particular $[m_{(2)}] = [m_{(1,1)}] = n+1$, so these coefficients do not stabilize.

4 Finite affine, Lie-type, and exceptional finite families

4.1 Power characters, parabolic cycles, and bounded flags

For every honest complex representation of a finite group, character values on powers give the exact class-compressed formula

$$\boxed{\mathcal{S}_{G,V} = \sum_{[g]\subseteq G} \frac{1}{|C_G(g)|} \exp\left(\sum_{r\ge 1} \frac{\chi_V(g^r)}{r} p_r \right).} \tag{14}$$

Thus the conjugacy classes, power maps, and character table determine the entire Lambda-distribution. In particular,

$$\boxed{[m_\tau]\mathcal{S}_{G,V} = \frac{1}{|G|} \sum_{g\in G} \prod_j \left(\sum_{\lambda \vdash \tau_j} \frac{1}{z_\lambda} \prod_{r\ge 1} \chi_V(g^r)^{m_r(\lambda)} \right).} \tag{15}$$

For the coset action $X = G/H$, fixed points and cycles are

$$\boxed{F_d(g; G/H) = \frac{|C_G(g^d)|}{|H|} |(g^d)^G \cap H|, \qquad c_r(g; G/H) = \frac{1}{r} \sum_{d|r} \mu(r/d) F_d(g; G/H).} \tag{16}$$

Substitution in (10) is an exact formula for every finite Lie-type parabolic action G/P, including projective, Grassmann, polar, and fixed-shape partial-flag actions. For conjugation on G,

$$\boxed{c_r^{\mathrm{conj}}(g) = \frac{1}{r} \sum_{d|r} \mu(r/d) |C_G(g^d)|, \qquad \mathcal{S}_{G,\mathrm{conj}} = \frac{1}{|G|} \sum_g \prod_{i,r} (1-t_i^r)^{-c_r^{\mathrm{conj}}(g)}.} \tag{17}$$

Fix q and a flag shape $\mathbf{d} = (d_1 < \cdots < d_s)$ with $d = d_s$. For $X_n = \mathrm{Flag}_{\mathbf{d}}(\mathbb{F}_q^n)$ and $D = |\tau|$,

$$\boxed{[m_\tau]\mathcal{S}_{GL_n(q),X_n} \text{ is independent of } n \text{ for } n \ge dD.} \tag{18}$$

The same range holds for $PGL_n(q)$; $SL_n(q)$ and $PSL_n(q)$ have the safe range $n > dD$. The stable coefficient is the sum, over $r \le dD$, of $GL_r(q)$-orbits of spanning τ-colored flag multisets. Fixed-shape isotropic and singular flags in each symplectic, unitary, or separate orthogonal tower also stabilize after a finite tower-dependent threshold, with the restricted form included in the supported configuration data. Complete flags are excluded: $[m_{(1,1)}] = |W|$ already grows with rank.

Steinberg and specified genuine Weil representations are obtained by inserting their characters in (14). A Deligne–Lusztig virtual character gives a completed Grothendieck-ring series whose coefficients can be virtual. A defining-characteristic linear action on $\mathfrak{g}(\mathbb{F}_q)$ is not automatically a complex representation; its finite-set conjugation action is an honest permutation alternative.

Exact verification used the standard/Steinberg module of $S_3 \cong GL_2(2)$, all 168 matrices of $GL_3(2)$ on G/P, conjugation by S_3, and consecutive stable ranks for projective points and 2-planes. All 26 coefficient and stability rows passed; the detailed proof and marimo notebook are included in the proof anthology.

4.2 Affine and finite classical actions

For $\mathrm{AGL}_n(q)$ on $\mathbb{F}_q^n$,

$$\boxed{\mathcal{S}_{n,q} = \frac{1}{q^n|GL_n(q)|}\sum_{A,b}\prod_r Q_r(\mathbf{t})^{c_r(A,b)}, \qquad Q_r = \prod_i (1-t_i^r)^{-1}.} \tag{19}$$

The fixed-point calculation is the linear system $(I-A^r)x = (I+A+\cdots+A^{r-1})b$; coefficients stabilize for $n \geq |\tau|-1$.

For nonzero vectors and projective points of finite linear groups,

$$|\operatorname{Fix}_{\mathbb{F}_q^n\setminus 0}(g^r)| = q^{\dim\ker(g^r-I)}-1, \quad |\operatorname{Fix}_{\mathbb{P}^{n-1}}(g^r)| = \sum_{\lambda\in\mathbb{F}_q^\times}\frac{q^{\dim\ker(g^r-\lambda I)}-1}{q-1}.$$

For complete flags, $[m_{(1,1)}] = |W|$, so rank stability already fails in degree two. For stable polar point actions, $[m_{(2)}] = [m_{(1,1)}] = 3$ (coincident, perpendicular, nonperpendicular).

4.3 $PSL_2(q)$, Δ groups, and code-monomial groups

For $PSL_2(q)$ on $\mathbb{P}^1(\mathbb{F}_q)$, grouping identity, unipotent, split, and nonsplit semisimple elements gives

$$\begin{aligned}\mathcal{S} = \frac{1}{|G|}\Big[&\operatorname{Exp}_\sigma((q+1)p_1) + (q^2-1)\operatorname{Exp}_\sigma(p_1+(q/p)p_p)\\ &+\frac{q(q+1)}{2}\sum_{\substack{m|(q-1)/d\\ m>1}}\varphi(m)\operatorname{Exp}_\sigma(2p_1+((q-1)/m)p_m)\\ &+\frac{q(q-1)}{2}\sum_{\substack{m|(q+1)/d\\ m>1}}\varphi(m)\operatorname{Exp}_\sigma(((q+1)/m)p_m)\Big],\end{aligned}$$

where $d=\gcd(2,q-1)$. The degree-three coefficient is $4+d$; the cross-ratio term makes degree four grow with q.

For the standard $SU(3)$ monomial groups,

$$\boxed{\mathcal{S}_{\Delta(3n^2)} = \tfrac{1}{3}\mathcal{A}_n + \tfrac{2}{3}\operatorname{Exp}_\sigma(p_3), \quad \mathcal{S}_{\Delta(6n^2)} = \tfrac{1}{6}\mathcal{A}_n + \tfrac{1}{3}\operatorname{Exp}_\sigma(p_3) + \tfrac{1}{2}\mathcal{B}_n.} \tag{20}$$

Here $\mathcal{A}_n$ averages the diagonal torus N_n, while $\mathcal{B}_n$ is the root-of-unity average of the transposition-coset factor $((1-\eta^a t_i^2)(1+\eta^{-a}t_i))^{-1}$.

The code-monomial family $D(C)\rtimes H$ obeys the dual-code cycle rule described after (6). Explicit groups of orders $8, 24, 18$ gave 57 exact comparisons. The Δ families gave 240 comparisons, and the $PSL_2(q)$ actions gave 48.

5 Braid, Pauli, extraspecial, Clifford, and Barnes–Wall groups

5.1 Finite braid images

The verified images are scalar cyclic quotients of B_3, the permutation quotient $B_3 \twoheadrightarrow S_3$, and the finite Ising/Jones image. For a scalar cyclic image C_m on dimension d,

$$[m_\tau]\mathcal{S} = \mathbf{1}_{m||\tau|}\prod_i\binom{d+\tau_i-1}{\tau_i}.$$

For the S_3 permutation image, the three class spectra give the exact $1/6$ weighted identity/transposition/three-cycle formula. The Ising lift has an eighth-root scalar selection rule; the projective adjoint removes scalar dependence and is checked separately.

5.2 Qudit Pauli and scalar-extended Pauli groups

Let $q = p^f$ with odd p, $N = q^n$, and define

$$A_\tau = \prod_i \binom{N + \tau_i - 1}{\tau_i}, \qquad B_\tau = \mathbf{1}_{p \mid \tau_i\ \forall i} \prod_i \binom{N/p + \tau_i/p - 1}{\tau_i/p}.$$

For the finite Pauli image,

$$\boxed{[m_\tau]\mathcal{S}_{P_{q,n}} = \begin{cases} 0, & p \nmid |\tau|, \\ q^{-2n} A_\tau + (1 - q^{-2n}) B_\tau, & p \mid |\tau|. \end{cases}} \tag{21}$$

For $\widetilde{P}^{(s)}_{q,n} = \mu_s P_{q,n}$, replace the selection rule by $s \mid |\tau|$. The noncentral term has a *plus* sign; the proposed qubit minus sign was disproved by the nonintegral coefficient it produced.

5.3 Extraspecial and Clifford groups

For an odd-p extraspecial group of exponent p, the Pauli formula (21) applies. Exponent p^2 splits the noncentral sector according to $g^p \in Z(E)$. For extraspecial 2-groups of type $\varepsilon \in \{+1, -1\}$, $N = 2^n$,

$$\begin{aligned} \mathcal{S}_{E^\varepsilon} = {} & 2^{-2n} \Pi_2 \prod_j (1 - t_j)^{-N} \\ & + (\tfrac{1}{2} + \varepsilon 2^{-n-1} - 2^{-2n}) \prod_j (1 - t_j^2)^{-N/2} + (\tfrac{1}{2} - \varepsilon 2^{-n-1}) \prod_j (1 + t_j^2)^{-N/2}. \end{aligned}$$

This distinguishes D_8 from Q_8.

For a Clifford group containing all scalars, the one-sided series is exactly 1. The intrinsic object is the phase-neutral two-sided series

$$[m_\alpha(\mathbf{s}) m_\beta(\mathbf{t})] \mathcal{B}^0_G = \dim(\operatorname{Sym}^\alpha V \otimes \operatorname{Sym}^\beta V^*)^G, \qquad |\alpha| = |\beta|.$$

Its $(1^k), (1^k)$ coefficient is the frame potential $|G|^{-1} \sum_g |\operatorname{Tr} g|^{2k}$, so

$$G \text{ is a unitary } k\text{-design} \iff \Phi_k(G) = \sum_{\lambda \vdash k,\ \ell(\lambda) \le d} (f^\lambda)^2.$$

5.4 Real Clifford and Barnes–Wall layers

For the real extraspecial Pauli group E_m with $N = 2^m$,

$$\boxed{\begin{aligned} \mathcal{S}_{E_m} = \frac{1}{2N^2} \Big[& \prod_j (1 - t_j)^{-N} + \prod_j (1 + t_j)^{-N} \\ & + (N^2 + N - 2) \prod_j (1 - t_j^2)^{-N/2} \\ & + (N^2 - N) \prod_j (1 + t_j^2)^{-N/2} \Big]. \end{aligned}} \tag{22}$$

For $\mathcal{C}_m = N_{O(2^m)}(E_m)$,

$$\mathcal{S}_{\mathcal{C}_m} = \frac{1}{|O^+(2m,2)|} \sum_{A \in O^+(2m,2)} \frac{1}{|E_m|} \sum_{e \in E_m} \prod_j \det(I - t_j e \widetilde{A})^{-1}.$$

The one-row stable coefficient $[m_{(2k)}]$ counts binary self-dual codes of length $2k$ up to isomorphism. For the Barnes–Wall lattice group $B_m \subset \mathcal{C}_m$ of index two,

$$\boxed{\mathcal{S}_{B_m} = \mathcal{S}_{\mathcal{C}_m} + \mathcal{S}^{\varepsilon_m}_{\mathcal{C}_m}.} \tag{23}$$

The twisted term is the lattice-sensitive sector.

6 Compact classical, exceptional, and Schur-functor families

6.1 Defining representations of $SU(n)$, $SO(n)$, $O(n)$, and tori

For the defining $SU(n)$ module,

$$\dim W_\tau^{SU(n)} = \begin{cases} K_{(q^n),\tau}, & |\tau| = nq, \\ 0, & n \nmid |\tau|. \end{cases}$$

For orthogonal groups,

$$\dim W_\tau^{O(n)} = \sum_{\ell(2\delta)\le n} K_{2\delta,\tau}, \qquad \dim W_\tau^{SO(n)} = \dim W_\tau^{O(n)} + \sum_{\ell(\delta)\le n} K_{2\delta+(1^n),\tau}.$$

The row-length restriction is essential. For a weighted torus T^r,

$$\mathcal{S}_{T^r,V} = \mathrm{CT}_{z_1,\dots,z_r} \prod_{i,j} (1 - t_i z^{w_j})^{-1}.$$

6.2 Exceptional and adjoint compact groups

All use the exact Weyl constant term (3). Their verified low-degree signatures are

$$\begin{aligned}
\mathcal{S}_{G_2,7} &= 1 + h_2 + e_3 + O_{\ge 4}, \\
\mathcal{S}_{\mathrm{Spin}(7),8} &= 1 + h_2 + h_4 + s_{(2,2)} + e_4 + O_{\ge 6}, \\
\mathcal{S}_{F_4,26} &= 1 + h_2 + h_3 + O_{\ge 4}, \\
\mathcal{S}_{E_6,27} &= 1 + h_3 + O_{\ge 6}, \quad \mathcal{S}(t) = (1-t^3)^{-1}, \\
\mathcal{S}_{E_7,56} &= 1 + e_2 + h_4 + s_{(2,2)} + e_4 + O_{\ge 6}, \quad \mathcal{S}(t) = (1-t^4)^{-1}, \\
\mathcal{S}_{E_8,248} &= 1 + h_2 + e_3 + O_{\ge 4}, \\
\mathcal{S}_{SU(2),\mathrm{Ad}} &= 1 + h_2 + e_3 + O_{\ge 4}, \\
\mathcal{S}_{SU(n),\mathrm{Ad}} &= 1 + h_2 + h_3 + e_3 + O_{\ge 4} \quad (n \ge 3).
\end{aligned}$$

The E_6 center imposes $3 \mid |\tau|$; the centers of E_7 and Spin(7) kill odd total degree. For E_8 the one-copy degrees are $2, 8, 12, 14, 18, 20, 24, 30$.

6.3 Schur functors of $U(n)$, $O(n)$, and $USp(2n)$

For every partition $\lambda \vdash d$,

$$\boxed{\mathrm{Tr}(S_\lambda(g)^r) = s_\lambda[X(g^r)] = \sum_{\alpha\vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a\in\alpha} \mathrm{Tr}(g^{ra}).} \tag{24}$$

For nonempty λ, the purely holomorphic $U(n)$ series is 1; the nontrivial series pairs the module with its conjugate. The stable orthogonal and symplectic coefficients are the plethystic pushforwards

$$[m_\tau]\mathcal{S}^O_{\lambda,\infty} = \langle \mathrm{Exp}_\sigma(h_2), h_\tau[s_\lambda] \rangle, \qquad [m_\tau]\mathcal{S}^{Sp}_{\lambda,\infty} = \langle \mathrm{Exp}_\sigma(e_2), h_\tau[s_\lambda] \rangle.$$

In both cases $d|\tau|$ odd forces zero. The trace limits are polynomial substitutions in Gaussian trace variables: circular complex for $U(n)$, $\sqrt{k}A_k + \mathbf{1}_{2|k}$ for $O(n)$, and $\sqrt{k}A_k - \mathbf{1}_{2|k}$ for $USp(2n)$.

6.4 Local unitary tensor products and conjugation

For $G = \prod_{a=1}^k U(d_a)$ on $V = \bigotimes_a \mathbb{C}^{d_a}$, the one-sided series is 1. The balanced coefficient vanishes unless $|\alpha| = |\beta|$, and in stable bidegree

$$\boxed{\mathcal{B}_{G,V} \equiv \sum_\lambda z_\lambda^{k-2} p_\lambda(\mathbf{s}) p_\lambda(\mathbf{t}).} \tag{25}$$

Exactly, $b_{(m),(m)}$ is the sum of squares of generalized Kronecker coefficients; for two factors it counts partitions of m obeying the dimension cutoffs.

For conjugation of $U(n)$ on $\mathrm{End}(\mathbb{C}^n)$,

$$\boxed{\mathcal{S} = \int_{U(n)} \exp\left(\sum_{k\geq 1} \frac{p_k}{k} |\operatorname{Tr}(g^k)|^2\right) dg.} \tag{26}$$

Finite coefficients are sums of squares of Littlewood–Richardson multiplicities. For $n \geq |\tau|$ they count H_τ-conjugacy orbits in $S_{|\tau|}$, equivalently multisets of colored necklaces, giving the stable Euler product $\prod_N (1 - \mathrm{wt}\, N)^{-1}$.

6.5 Proctor's odd symplectic group

The complex stabilizer is $W^* \rtimes (Sp_{2n}(\mathbb{C}) \times \mathbb{C}^\times)$ and is noncompact, so a normalized Haar expectation does not exist. Its full algebraic covariant-invariant series is 1. The maximal compact replacements are

$$\mathcal{S}_{Sp(n)\times U(1),\, W\oplus\chi} = \mathcal{S}_{Sp(n),W}, \qquad \mathcal{S}_{Sp(n),\, W\oplus\mathbf{1}} = \mathcal{S}_{Sp(n),W} \prod_i (1-t_i)^{-1}.$$

Over $\mathbb{F}_q$, the honest complex permutation module on $\mathbb{P}(W \oplus L)$ obeys (10); it has two point orbits, so $[m_{(1)}] = 2$.

7 Polyhedral, sporadic, and special 24-dimensional groups

For a three-dimensional rotation through θ, put

$$\Phi_\theta = \prod_i \left((1-t_i)(1-2\cos\theta\, t_i + t_i^2)\right)^{-1}.$$

The A_4, S_4, A_5 rotation series are the class-weighted averages of Φ_θ with their usual angle counts. Adjoining negatives gives the H_3 reflection group; its one-variable Molien series is $((1-t^2)(1-t^6)(1-t^{10}))^{-1}$. The binary lifts $2T, 2O, 2I \leq SU(2)$ replace θ by paired half-angle spectra; $-I$ kills all odd degrees. The tested A_5 modules include the permutation, deleted-permutation, and three-dimensional rotation representations.

For the M_{24} permutation action, if a class has cycle shape $\prod_r r^{c_r(C)}$,

$$\boxed{\mathcal{S}_{M_{24}} = \sum_C \frac{|C|}{|M_{24}|} \prod_{i,r} (1-t_i^r)^{-c_r(C)}.}$$

Five-transitivity gives agreement with S_{24} through degree five; the first extra orbit occurs at degree six. For the Leech representation of Co_0 with frame shape $\prod_r r^{k_r(C)}$,

$$\boxed{\mathcal{S}_{Co_0} = \sum_C \frac{|C|}{|Co_0|} \prod_{i,r} (1-t_i^r)^{-k_r(C)}, \qquad [m_\tau] = 0 \text{ for odd } |\tau|.}$$

The signed-coordinate subgroup checks are not promoted to full-Co_0 coefficients.

8 Nonuniform measures, heat flow, and limiting laws

For a finite walk $\nu = \kappa^{*r}$, representation Fourier transforms obey $\widehat{\nu}(W) = \widehat{\kappa}(W)^r$. For a cyclic diagonal representation,

$$[m_\tau]\mathcal{S}_{\nu,V} = \sum_{q \bmod m} M_\tau(q)\widehat{\nu}(q).$$

For a probability measure on S_n, the permutation series is the transformed cycle-count PGF $\mathbb{E}_\nu \prod_d Q_d(\mathbf{t})^{C_d}$. This covers random transpositions, random 3-cycles, adjacent transpositions, riffle shuffles, Ewens measures, and general cycle weights. For Ewens parameter θ,

$$\mathcal{S}_{n,\theta} = \frac{[z^n] \exp(\theta \sum_{d\geq 1} Q_d z^d/d)}{[z^n] \exp(\theta \sum_{d\geq 1} z^d/d)}.$$

For $U(1)$ heat time t and a weight decomposition $W_\tau = \bigoplus_w M_\tau(w)\mathbb{C}_w$,

$$\boxed{[m_\tau]\mathcal{S}_t = \sum_{w\in\mathbb{Z}} M_\tau(w)e^{-w^2t/2}.} \tag{27}$$

This interpolates from dimension at $t = 0$ to Haar invariants as $t \to \infty$.

For fixed k, the S_n action on k-subsets has the fixed-power formula (12). With independent $Z_j \sim \mathrm{Poisson}(1/j)$,

$$F_{k,r}^{(\infty)} = [u^k]\prod_{j=1}^{kr}(1 + u^{j/\gcd(j,r)})^{\gcd(j,r)Z_j}, \quad \mathcal{S}_\infty^{(k)} = \sum_\lambda \frac{\mathbb{E}\prod_a F_{k,\lambda_a}^{(\infty)}}{z_\lambda} p_\lambda.$$

For $k \geq 2$ this limit is non-Poisson; its variance is k.

For a rooted tree whose branch isomorphism types A occur with multiplicity m_A,

$$\boxed{\mathcal{S}_T(\mathbf{t}) = \prod_A [u_A^{m_A}] \exp\left(\sum_{\ell\geq 1} \frac{u_A^\ell}{\ell}\mathcal{S}_A(\mathbf{t}^\ell)\right).} \tag{28}$$

This is exact for every finite tree. A general random-tree Lambda limit remains a separate probabilistic problem.

9 New-distribution formula and asymptotic ledger

This section indexes every canonical note merged from `proved_matrix_groups/anthology_sections/new_dists/`. Each displayed expression is an explicit sigma-MGF (or the explicitly marked projective/balanced replacement), rather than only a coefficient recipe. The full hypotheses, proofs, exceptional cases, and verification tables appear in the final part of the proof anthology.

9.1 Association-scheme permutation groups

For any of the Johnson, Hamming, Doob, Grassmann, polar, building, translation, or distance-regular actions $G \curvearrowright X$, define

$$F_r(g) = |\operatorname{Fix}_X(g^r)|, \qquad c_d(g) = \frac{1}{d}\sum_{r|d}\mu(d/r)F_r(g).$$

Then the explicit group/action series is

$$\boxed{\mathcal{S}_{G,X} = \frac{1}{|G|}\sum_{g\in G}\prod_{i,d\geq 1}(1 - t_i^d)^{-c_d(g)}.} \tag{29}$$

For example, if $\lambda = 1^{m_1}2^{m_2}\cdots \vdash v$,

$$F_r^{J(v,k)}(\lambda) = [u^k]\prod_{j\geq 1}(1 + u^{j/\gcd(j,r)})^{m_j \gcd(j,r)}, \quad \mathcal{S}_{J(v,k)} = \sum_{\lambda\vdash v}\frac{1}{z_\lambda}\prod_{i,d}(1 - t_i^d)^{-c_d^J(\lambda)}.$$

Asymptotics. For fixed k and total degree D, Johnson and Grassmann coefficients stabilize once the ambient rank is at least kD. Hamming coefficients stabilize in the alphabet size for $q \geq D$. If both geometric parameters grow, pair-orbit counts such as $\min(k, v - k) + 1$, $d + 1$, or $|W|$ can diverge, obstructing a raw coefficientwise limit.

9.2 Primitive-action families

For an almost-simple coset action G/M, the substitution

$$F_r(g) = \frac{|C_G(g^r)|\,|(g^r)^G \cap M|}{|M|}$$

in (29) is the explicit MGF. For diagonal, product, twisted-wreath, holomorph-simple, and holomorph-compound actions, the same series holds with the family-specific fixed-power products recorded in the proof anthology. In particular, for $H \wr S_n$ on Ω^n,

$$F_r(h;\pi) = \prod_{C\in\mathrm{Cyc}(\pi)} |\,\mathrm{Fix}_\Omega(h_C^{r/e_C})|^{e_C}, \qquad e_C = \gcd(|C|, r).$$

Asymptotics. Almost-simple sequences require family data; product-action pair moments grow polynomially, while diagonal-type moments grow by centralizer powers. These are generally multiset or critical-branching limits, not a universal ordinary compound-Poisson law.

9.3 Finite local-ring actions

Let $R_a = \mathcal{O}/\pi^a\mathcal{O}$ with residue field $\mathbb{F}_q$, and let $K_b(A)$ be the kernel size of A over R_b. The primitive-vector and projective-point series for $\mathrm{GL}_n(R_a)$ are

$$\begin{aligned}
\mathcal{S}^{\mathrm{prim}}_{n,a} &= \frac{1}{|\,\mathrm{GL}_n(R_a)|}\sum_h \prod_{i,r}(1-t_i^r)^{-\frac{1}{r}\sum_{d\mid r}\mu(r/d)[K_a(h^d-I)-K_{a-1}(h^d-I)]},\\
\mathcal{S}^{\mathrm{proj}}_{n,a} &= \frac{1}{|\,\mathrm{GL}_n(R_a)|}\sum_h \prod_{i,r}(1-t_i^r)^{-\frac{1}{r}\sum_{d\mid r}\mu(r/d)F_d^{\mathrm{proj}}(h)},\\
F_d^{\mathrm{proj}}(h) &= \frac{1}{|R_a^\times|}\sum_{u\in R_a^\times}[K_a(h^d-uI)-K_{a-1}(h^d-uI)].
\end{aligned}$$

Asymptotics. Fixed level and bounded degree give rank stability, but level growth does not: $[m_{(1,1)}]\mathcal{S}^{\mathrm{prim}} = q^a$ and $[m_{(1,1)}]\mathcal{S}^{\mathrm{proj}} = a+1$ in stable rank.

9.4 Code-monomial matrix groups

For $C \le (\mathbb{Z}/r)^N$, $P \le \mathrm{Aut}_{\mathrm{perm}}(C)$, and $G(C,P) = D(C) \rtimes P$, write

$$H_a(\mathbf{t}) = \frac{1}{r}\sum_{s=0}^{r-1}\zeta^{-as}\prod_i(1-\zeta^s t_i)^{-1}.$$

The explicit weighted-cycle and dual-code forms are

$$\boxed{\mathcal{S}_{C,P} = \frac{1}{|C||P|}\sum_{\pi,c}\prod_{i,Q\in\mathrm{Cyc}(\pi)}(1-\zeta^{\sum_{j\in Q}c_j}t_i^{|Q|})^{-1} = \frac{1}{|P|}\sum_\pi \sum_{\substack{u\in C^\perp\\ u\text{ constant on }Q}}\prod_Q H_{u_Q}(\mathbf{t}^{|Q|}).} \tag{30}$$

For $P = 1$, this is the complete weight enumerator $\mathcal{S}_{D(C)} = \mathrm{cwe}_{C^\perp}(H_0,\dots,H_{r-1})$.

Asymptotics. Length-growing families stabilize only under bounded-support orbit and dual-code conditions. Pure diagonal families already have $[m_r]\mathcal{S} \ge N$, and many semidirect families have a diverging $[m_{r,r}]$ coefficient.

9.5 Multiplicative wreath and tensor-induced representations

For $G_n = H \wr S_n$, an H-wreath type $\boldsymbol{\mu}$, and its power transform $\boldsymbol{\mu}^{[r]}$,

$$\boxed{\mathcal{S}_{G_n,W} = \sum_{\boldsymbol{\mu}\vdash_H n}\frac{1}{Z_{\boldsymbol{\mu}}}\exp\left(\sum_{r\ge 1}\frac{p_r}{r}\chi_W(\boldsymbol{\mu}^{[r]})\right).} \tag{31}$$

For the product action on Ω^n, use the fixed-power product in the preceding subsection and (29). For $V^{\otimes n}$,

$$\chi(g^r) = \prod_{Q \in \mathrm{Cyc}(\pi)} \chi_V(h_Q^{r/e_Q})^{e_Q}, \qquad e_Q = \gcd(|Q|, r),$$

inside (31).

Asymptotics. Additive block modules have compound-Poisson limits. Product and tensor modules usually do not: if $R_s = \#(H\backslash\Omega^s)$ then $[m_{(1^s)}] = \binom{n+R_s-1}{R_s-1}$, and the analogous tensor coefficient uses $a_s = \dim(V^{\otimes s})^H$.

9.6 Central extensions and projective representations

If $1 \to Z \to \widetilde{G} \to G \to 1$ and Z acts through μ_r, then

$$\boxed{\mathcal{S}_{\widetilde{G},V} = \frac{1}{|G|}\sum_{g\in G}\frac{1}{r}\sum_{\zeta\in\mu_r}\prod_i \det(I-\zeta t_i\rho(\widetilde{g}))^{-1}, \quad [m_\tau] = 0 \text{ unless } r \mid |\tau|.} \tag{32}$$

For a genuinely projective representation, the lift-independent object is

$$\boxed{\mathcal{B}^{\mathrm{proj}}_{G,[V]}(\mathbf{s},\mathbf{t}) = \mathrm{CT}_u \int_G \prod_i \det(I-us_iU_g)^{-1}\prod_j \det(I-u^{-1}t_jU_g^{-*})^{-1}\,dg.}$$

This covers Schur/spin covers, extraspecial central products, Weil lifts, Clifford groups, determinant-one lifts, and sporadic covers.

Asymptotics. Finite scalar order gives congruence sectors; a continuous scalar center makes the one-sided series equal to 1. Balanced design moments may stabilize, while one-sided lift-dependent coefficients need not.

9.7 Braid, TQFT, and quantum-image groups

For a finite braid image I with represented character χ,

$$\boxed{\mathcal{S}_{I,V} = \sum_{[x]\subseteq I}\frac{1}{|C_I(x)|}\exp\left(\sum_{r\geq 1}\frac{p_r}{r}\chi(x^r)\right).}$$

For a monomial image $x = D(\alpha)P_\pi$, this becomes $|I|^{-1}\sum_x \prod_{i,C}(1-\alpha_C(x)t_i^{|C|})^{-1}$. Infinite images use the normalized Haar measure on the compact closure, not a finite-image average. The Ising lift has order 192 and scalar center μ_8, so only total degrees divisible by eight survive.

Asymptotics. Dense $U(d)$ closure has one-sided series 1; stable orthogonal and symplectic closures are governed by contraction algebras. Varying finite images or fusion dimensions have no representation-free limit.

9.8 Lattice automorphism groups

For a positive-definite lattice L, write $\det(I-tg) = \prod_m (1-t^m)^{a_m(g)}$. Then

$$\boxed{\mathcal{S}_L = \sum_{[g]\subseteq \mathrm{Aut}(L)}\frac{1}{|C_{\mathrm{Aut}(L)}(g)|}\prod_{i,m}(1-t_i^m)^{-a_m(g)}.} \tag{33}$$

Since $-I \in \mathrm{Aut}(L)$, every odd-total-degree coefficient vanishes.

Asymptotics. Type A has a Poisson-cycle limit; signed-permutation $B/C/D$ towers have a stable signed-cycle law; repeated orthogonal sums have a marked compound-Poisson limit. Niemeier and isolated extremal/modular lattices are finite collections, not intrinsic rank towers.

9.9 Sporadic groups: the explicit M_{11} series

With $Z_\lambda = \prod_{i,r}(1-t_i^r)^{-c_r(\lambda)}$,

$$\boxed{\begin{aligned}\mathcal{S}_{M_{11},\mathbb{C}^{11}} = \frac{1}{7920}\big(&Z_{1^{11}} + 165Z_{1^3\,2^4} + 440Z_{1^2\,3^3} + 990Z_{1^3\,4^2}\\ &+ 1584Z_{1\,5^2} + 1320Z_{2\,3\,6} + 1980Z_{1\,2\,8} + 1440Z_{11}\big).\end{aligned}}$$

The deleted permutation module W has the explicit series

$$\mathcal{S}_{M_{11},W} = \prod_i (1-t_i)\mathcal{S}_{M_{11},\mathbb{C}^{11}}.$$

Asymptotics. There is no canonical sequence through the sporadic groups. For a fixed representation the formula is a rational function and ordinary coefficient asymptotics apply; covers and other modules require their own represented character and power maps.

9.10 Solvable and finite-subgroup families

For the natural monomial lamplighter representation $L_{r,n} = \mu_r^n \rtimes C_n$, let $A_{r,\ell} = r^{-1}\sum_{\zeta^r=1}\prod_i(1-\zeta t_i^\ell)^{-1}$. Then

$$\boxed{\mathcal{S}^{\mathrm{mon}}_{L_{r,n}} = \frac{1}{n}\sum_{\ell\mid n}\varphi(\ell)A_{r,\ell}^{n/\ell}.}$$

For $M(m,n,u) = \langle a,b \mid a^m = b^n = 1,\ bab^{-1} = a^u\rangle$ in its induced monomial representation,

$$\boxed{\mathcal{S}_{M(m,n,u)} = \frac{1}{mn}\sum_{x,y}\prod_{i,C\in\mathrm{Cyc}(y)}(1-\alpha_C(x,y)t_i^{|C|})^{-1}.}$$

The affine lamplighter, Frobenius, unitriangular/Borel/parabolic, finite unitary/symplectic, and reflection-wreath cases use (29), the class-power formula, or (7) according to the stated representation.

Asymptotics. The monomial lamplighter has a linearly growing $[m_{r,r}]$ coefficient; the affine action has necklace growth asymptotic to r^n/n. Reflection wreath blocks have a genuine compound-Poisson limit; isolated finite subgroups do not form a canonical rank tower.

9.11 Compact Spin and Pin groups

For the type-B_n spin module Δ_n,

$$\boxed{\mathcal{S}_{\mathrm{Spin}(2n+1),\Delta_n} = \frac{1}{2^n n!}\mathrm{CT}_z\left[\mathcal{D}_{B_n}(z)\exp\left(\sum_{r\ge1}\frac{p_r}{r}\prod_{j=1}^n(z_j^{r/2}+z_j^{-r/2})\right)\right].} \tag{34}$$

Type D_n half-spin replaces the B_n denominator and character by the parity-selected half-spin weights. $\mathrm{Pin}^{\pm}(N)$ is the half-sum of the Spin integral and the appropriate odd-component integral.

Asymptotics. $\dim(\Delta_n^{\otimes4})^{\mathrm{Spin}(2n+1)} = n+1$; half-spin fourth moments also grow linearly by parity. Hence the unnormalized spinor and half-spin sigma-MGFs have no finite coefficientwise large-rank limit.

9.12 Non-Haar Fourier-weighted laws

For an arbitrary probability measure ν on a finite or compact group,

$$\boxed{\mathcal{S}_{\nu,V} = \int_G \exp\left(\sum_{r\geq 1} \frac{p_r}{r}\chi_V(g^r)\right) d\nu(g), \quad [m_\tau]\mathcal{S}_{\nu^{*k},V} = \sum_{\pi\in\widehat{G}} m_\pi(W_\tau)\,\mathrm{Tr}(\widehat{\nu}(\pi)^k).} \tag{35}$$

For a central walk this reduces to $\sum_\pi m_\pi(W_\tau) d_\pi \theta_\pi^k$.

Asymptotics. For fixed G, the largest nontrivial Fourier eigenvalue visible in W_τ gives the exact leading approach to Haar. Heat kernels replace it by $e^{-t\kappa_\pi/2}$; growing-rank cutoff claims require a fixed metric normalization.

9.13 Symmetric, spin-cover, Lie, Foulkes, and tree families

For an S_n-module with Frobenius characteristic f_n,

$$\boxed{\mathcal{S}_{V_n} = \sum_{\mu\vdash n} \frac{1}{z_\mu} \exp\left(\sum_{r\geq 1} \frac{p_r}{r} z_{\mu^{[r]}}[p_{\mu^{[r]}}]f_n\right).} \tag{36}$$

This specializes to Specht, Schur-cover genuine, Foulkes, and free-Lie modules (for the latter, $f_n = n^{-1}\sum_{d|n}\mu(d)p_d^{n/d}$). Rooted-tree leaf actions use the exact recursive cycle polynomial and the general permutation MGF (29).

Asymptotics. Fixed-tail Specht modules have polynomial Poisson-chaos limits. Growing Foulkes shapes, free-Lie modules, and level-transitive rooted-tree actions have diverging low moments and no raw coefficientwise limit; arbitrary random trees need a separately specified probabilistic model.

10 Verification ledger and corrections

Family	Independent evidence	Reported checks
Association schemes	matrices, cycle products, geometric fixed points, orbit traversal	66
Matrix-group formula suite	compact kernels, finite Reynolds averages, spectral classes, dual-code counts	928
Schur functors $U/O/USp$	explicit Young-symmetrizer matrices versus plethystic traces	120
$\Delta(3n^2), \Delta(6n^2)$	explicit matrices versus Q_n/R_n coefficient formulas	240
Nonuniform S_n	direct matrices versus cycle transform and Fourier operators	267
$U(1)$ heat flow	quadrature versus weight/Casimir attenuation	285
Hyperoctahedral cube	explicit 2^n-dimensional matrices versus signed-cycle law	all $n \leq 5$
Rooted trees	recursive matrix closures versus wreath recurrence	30
Generalized dihedral	Reynolds ranks versus classified spectral formula	33
Odd symplectic	compact constant terms, vanishing certificate, finite permutation groups	76
Other packages	package-specific exact/numerical suites documented in their PDFs	pass

The consolidated work establishes the following corrections and boundaries:

- Orthogonal Schur shapes require $\ell(\lambda) \leq n$.
- A three-dimensional rotation factor is $(1-t)^{-1}(1-2\cos\theta\, t+t^2)^{-1}$.
- The scalar-extended qubit Pauli noncentral sector has a plus sign.
- One-sided holomorphic series collapse to 1 whenever a continuous scalar center acts nontrivially; balanced two-sided series carry the content.
- Proctor's complex odd symplectic group is noncompact and has no normalized Haar probability.
- Stable-range, finite-rank, numerical-Haar, and exact finite-group claims are kept distinct throughout.

Artifact map. Sources live in `lambda_distributions/proofs/`, `lambda_distributions/dists/`. All repository marimo laboratories are centralized and cataloged under `marimo_notebooks/`; the exact backends and pytest entry points remain beside their group-family sources. The legacy paths `lambda_distributions/proofs/` and `output/pdf/lambda_distributions/proofs/` are compatibility links. This compendium indexes, but does not replace, the detailed proofs.